

Stock Trading Simulator

OOP Project — Test Cases Report

Project	Stock Trading Simulator (PSX Market)
Course	Object Oriented Programming (OOP)
Language	C++17
Test File	test_cases.cpp
Total TCs	46
Final Result	46 PASSED 0 FAILED

Group 1 — Stock Construction & Getters [TC01–TC06]

ID	Test Name	Input / Condition	Result
TC01	Stock symbol getter	Stock s("OGDC", 200, 220, 225, 195, 210, 10, 5, 500000); s.getSymbol() == "OGDC"	PASS
TC02	Stock close price getter	s.getClose() == 210.0	PASS
TC03	Stock volume getter	s.getVolume() == 500000LL	PASS
TC04	Stock changePercent getter	s.getChangePercent() == 5.0	PASS
TC05	Stock high getter	s.getHigh() == 225.0	PASS
TC06	Stock low getter	s.getLow() == 195.0	PASS

Group 2 — Stock Comparison Operators [TC07–TC08]

ID	Test Name	Input / Condition	Result
TC07	operator> — higher changePct wins	Stock a(5% gain), b(2% gain) → a > b is true	PASS
TC08	operator< — lower changePct loses	b < a is true	PASS

Group 3 — Portfolio Initial State [TC09–TC12]

ID	Test Name	Input / Condition	Result
TC09	Initial cash balance is 100000	Portfolio p(100000.0); p.getCashBalance() == 100000.0	PASS
TC10	No shares held initially	p.getShares("OGDC") == 0	PASS
TC11	No invested amount initially	p.getInvestedAmount("OGDC") == 0.0	PASS
TC12	Total invested starts at zero	p.getTotalInvested() == 0.0	PASS

Group 4 — Buy Stock (Normal Flow) [TC13–TC16]

ID	Test Name	Input / Condition	Result
TC13	Cash decreases after buy	Buy 10 x PS0 @ 150 → <code>getCashBalance() == 98500.0</code>	PASS
TC14	Share count increases after buy	<code>getShares("PS0") == 10</code>	PASS
TC15	Trade history updated after buy	<code>getTradeHistory().size() == 1</code>	PASS
TC16	Last message reflects buy	<code>getLastMessage().find("PS0") != npos</code>	PASS

Group 5 — Sell Stock (Normal Flow) [TC17–TC20]

ID	Test Name	Input / Condition	Result
TC17	Cash increases after sell	Buy 20 x LUCK@200, sell 10 → <code>cash += 2000</code>	PASS
TC18	Share count decreases after sell	<code>getShares("LUCK") == 10</code>	PASS
TC19	Trade history has two records (buy + sell)	<code>getTradeHistory().size() == 2</code>	PASS
TC20	Last message reflects sell	<code>getLastMessage().find("LUCK") != npos</code>	PASS

Group 6 — Sell All Shares [TC21–TC22]

ID	Test Name	Input / Condition	Result
TC21	Selling all shares removes holding entry	Buy 5 then sell 5 ENGR0 → <code>getShares() == 0</code>	PASS
TC22	Cash returns to original after round-trip	<code>getCashBalance() == 100000.0</code>	PASS

Group 7 — Exception: InsufficientFunds [TC23–TC24]

ID	Test Name	Input / Condition	Result
TC23	InsufficientFundsException message not empty	Buy 2 x RICH@600000 (cost=120000 > balance=100000) → <code>e.what().size() > 0</code>	PASS
TC24	InsufficientFundsException is thrown	<code>caught == true</code>	PASS

Group 8 — Exception: InvalidQuantity on Buy (qty=0) [TC25]

ID	Test Name	Input / Condition	Result
TC25	InvalidQuantityException thrown on buy with qty=0	<code>buyStock(s, 0) → throws InvalidQuantityException</code>	PASS

Group 9 — Exception: InvalidQuantity on Buy (qty=-5) [TC26]

ID	Test Name	Input / Condition	Result
TC26	InvalidQuantityException thrown on buy with qty=-5	<code>buyStock(s, -5) → throws InvalidQuantityException</code>	PASS

Group 10 — Exception: InsufficientShares (oversell) [TC27–TC28]

ID	Test Name	Input / Condition	Result
TC27	InsufficientSharesException message not empty	Buy 5 UBL, <code>sellStock(s,10) → e.what().size() > 0</code>	PASS
TC28	InsufficientSharesException thrown when overselling	<code>caught == true</code>	PASS

Group 11 — Exception: Sell Stock Never Owned [TC29]

ID	Test Name	Input / Condition	Result
TC29	InsufficientSharesException thrown when selling unowned	<code>sellStock(MCB, 1)</code> without any prior buy → exception thrown	PASS

Group 12 — Exception: InvalidQuantity on Sell [TC30]

ID	Test Name	Input / Condition	Result
TC30	InvalidQuantityException thrown on sell with qty=0	<code>buyStock(s,5); sellStock(s,0) → throws InvalidQuantityException</code>	PASS

Group 13 — Exception Hierarchy (TradingException base) [TC31–TC32]

ID	Test Name	Input / Condition	Result
TC31	InsufficientFundsException caught as TradingException	<code>catch(const TradingException&) catches InsufficientFundsException</code>	PASS
TC32	InsufficientSharesException caught as TradingException	<code>catch(const TradingException&) catches InsufficientSharesException</code>	PASS

Group 14 — TradeRecord Fields [TC33–TC37]

ID	Test Name	Input / Condition	Result
TC33	TradeRecord action is BUY	<code>TradeRecord tr("BUY","OGDC",10,150,"..."); tr.getAction() == "BUY"</code>	PASS
TC34	TradeRecord symbol is OGDC	<code>tr.getSymbol() == "OGDC"</code>	PASS
TC35	TradeRecord quantity is 10	<code>tr.getQuantity() == 10</code>	PASS
TC36	TradeRecord total = qty * price	<code>tr.getTotalAmount() == 1500.0</code>	PASS
TC37	TradeRecord toCSV contains symbol	<code>tr.toCSV().find("OGDC") != npos</code>	PASS

Group 15 — Top Gainers Ordering [TC38–TC40]

ID	Test Name	Input / Condition	Result
TC38	getTopGainers returns exactly 2 stocks	<code>top.size() == 2</code>	PASS
TC39	Top gainer is the stock with highest %	<code>top[0].getSymbol() == "HIGH" (changePct=10%)</code>	PASS
TC40	Second gainer is correct	<code>top[1].getSymbol() == "MID" (changePct=3%)</code>	PASS

Group 16 — Merge Sort by Volume (Descending) [TC41–TC43]			
ID	Test Name	Input / Condition	Result
TC41	Merge sort: highest volume is first	After sort: <code>sorted[0].getVolume() == 1500</code>	PASS
TC42	Merge sort: middle volume is second	<code>sorted[1].getVolume() == 1000</code>	PASS
TC43	Merge sort: lowest volume is last	<code>sorted[2].getVolume() == 500</code>	PASS

Group 17 — Multiple Buys Accumulate [TC44–TC45]			
ID	Test Name	Input / Condition	Result
TC44	Multiple buys accumulate shares	Buy 10+5 HUBC → <code>getShares() == 15</code>	PASS
TC45	Cash deducted correctly after multiple buys	<code>getCashBalance() == 100000 - 750 == 99250.0</code>	PASS

Group 18 — Cash Balance Never Goes Negative [TC46]			
ID	Test Name	Input / Condition	Result
TC46	Cash balance never goes negative	Buy 100 x KAPC0@999 (OK), then buy 101 (fails) → <code>InsufficientFundsException</code> & <code>balance >= 0</code>	PASS

TOTAL TEST CASES	PASSED	FAILED	STATUS
46	46	0	ALL PASSED

Compile command: `g++ -std=c++17 test_cases.cpp src/*.cpp -I include -o test_runner && ./test_runner`